SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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SAVEETHA SCHOOL OF ENGINEERING

ASSESSMENT TOOL – 1(CO1)

ECA1308 – MICROWAVE ENGINEERING

APPLICATION-BASED ASSESSMENT

Analysis, Design and Verification of a Microwave Transmission Line and Network for a Selected Application

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REPORT SCOPE AND DESIGN BASIS

This report follows the seven assessment components and the submission requirements in the ECA13 Microwave Engineering CO1 assessment brief. The selected illustrative application is a 2.4 GHz wireless receiver feed using a nominal 50 Ω coaxial transmission line. The line dimensions and dielectric properties are modelling assumptions for analysis; they are not measurements from a manufactured cable.

The MATLAB script supplied for this report is included in Appendix A. Output spaces are provided for figures generated in MATLAB. Paste the corresponding images into the marked areas and retain the figure captions.

Important modelling boundary: the script estimates the TE11 upper-mode limit using an approximate expression and calculates distributed conductor/dielectric loss using a low-loss model. It does not by itself constitute a complete full-wave electromagnetic simulation or a measured hardware validation. Connector, transition, radiation and mismatch losses must be treated separately when finalising the loss budget.

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1. STAGE 1 – FORMULATE THE APPLICATION AND OPERATING REQUIREMENTS

1.1 Selected application

Application: a coaxial feed network connecting a 2.4 GHz wireless communication receiver to an RF front-end or antenna subsystem. Category: civil wireless communications. Coaxial cable is widely used for RF interconnections because its outer conductor provides shielding, its characteristic impedance can be controlled through conductor geometry and dielectric choice, and its TEM mode has no cutoff frequency.

The selected 2.4 GHz frequency lies in the globally used 2.4 GHz ISM allocation, although permitted channels, bandwidth and power limits depend on the jurisdiction and specific radio technology. A practical implementation must follow applicable local radio regulations and the device manufacturer's limits.

1.2 Brief historical context

Maxwell's electromagnetic theory unified electricity and magnetism and predicted electromagnetic waves. Hertz experimentally demonstrated electromagnetic-wave propagation. Microwave technology advanced significantly through cavity magnetrons and radar systems during the Second World War, followed by satellite communication, low-noise microwave receivers, microwave integrated circuits (MICs/MMICs), and modern Wi-Fi and cellular systems. Controlled transmission structures remain essential because they guide RF energy, limit unintended radiation and establish predictable impedance.

1.3 Frequency-band comparison

The following is a qualitative engineering comparison. Exact atmospheric attenuation and available bandwidth depend on frequency, weather, path length, regulation and system design.

Band	Representative frequency	Approx. free-space wavelength	Advantages	Limitations for this application
L band	1–2 GHz	150–300 mm	Lower free-space path loss and generally low rain sensitivity	Larger antennas and components; less aligned with a 2.4 GHz receiver
S band / 2.4 GHz ISM	2–4 GHz	75–150 mm; 2.4 GHz ≈ 125 mm in air	Compact RF components; mature Wi-Fi/ISM ecosystem; moderate propagation loss	Shared spectrum and possible interference from other devices
C band	4–8 GHz	37.5–75 mm	Smaller components and broader potential bandwidth	Higher free-space path loss than at lower frequencies; different hardware required
X band	8–12 GHz	25–37.5 mm	Compact high-frequency systems and radar applications	Greater path loss and tighter fabrication tolerances than 2.4 GHz
Ku/Ka bands	12–40 GHz approx.	25–7.5 mm approx.	Large bandwidth and compact apertures	Rain attenuation and high-frequency design complexity increase, especially at Ka band

1.4 Selected operating band

The model uses a centre frequency of 2.400 GHz and an illustrative analysis bandwidth of 100 MHz, giving a sweep from 2.350 GHz to 2.450 GHz. This bandwidth is an analysis choice for the assignment, not a claim that every 100 MHz channel is legally available for a specific device.

$f_0 = 2.400$ GHz; BW = 100 MHz; $f_1 = 2.350$ GHz; $f_2 = 2.450$ GHz

1.5 Quantified requirement specification

Parameter	Proposed design requirement	Basis / status
Application	2.4 GHz wireless receiver feed	Selected civil application
Operating band	2.350–2.450 GHz	Illustrative 100 MHz analysis band
Reference / characteristic impedance	50 Ω system; calculated line impedance reported separately	Standard RF reference; geometry gives calculated Zc
Maximum distributed line loss	Target ≤ 1.6 dB over 0.5 m for the preliminary model	Provisional target to be checked against the model
Return loss	Target ≥ 10 dB across the operating band	Design acceptance target
VSWR	Target ≤ 1.92	Equivalent to $\|\Gamma\| \approx 0.315$ at the 10 dB return-loss limit
Isolation / shielding	Specify from enclosure and system requirements; verify by EMC test	No shielding value is provided by the assessment brief
Power level	Low-power receiver feed; set the actual maximum from the selected radio hardware	No power rating assumed without component data
Size	L = 0.50 m; a = 0.50 mm; b = 1.70 mm in the model	Illustrative geometry
Cost	Use commercially available coax and connectors; enter a sourced local cost in final submission	No verified supplier quote used in this model

The impedance calculation produces approximately 48.95 Ω, which is close to but not exactly 50 Ω. This difference must be discussed rather than hidden. A final physical design could adjust the conductor ratio or dielectric constant to obtain a nominal 50 Ω line.

1.6 Constraints

- EMI/EMC: the coaxial outer conductor reduces external coupling and leakage when connectors, joints and terminations are properly installed.
- Safety: receiver-level signals are normally low power, but any transmitter or high-power test setup must follow RF exposure, electrical safety and equipment instructions.
- Environment: temperature and moisture can affect dielectric properties and cable loss; mechanical bending can change performance.
- Acceptance criteria: results must be checked over the complete 2.350–2.450 GHz band, not only at the centre frequency.

2. STAGE 2 – EVALUATE AND SELECT THE TRANSMISSION STRUCTURE

2.1 Mode and structure comparison

Structure	Principal mode	Cutoff behaviour	Loss / power / integration assessment
Coaxial line	TEM	TEM has no cutoff; higher-order modes impose an upper-frequency limit	Shielded, flexible, robust and easy to connect; conductor/dielectric loss depends on dimensions and materials
Rectangular waveguide	Dominant TE_{10}	Has a cutoff frequency; below cutoff, the dominant mode does not propagate	High power handling and potentially low loss at microwave frequencies; bulkier and requires transitions
Stripline	TEM (ideal homogeneous dielectric)	No TEM cutoff in the ideal model; practical higher-order limits remain	Good shielding and planar integration; fabrication and dielectric losses matter
Microstrip	Quasi-TEM	No simple TEM cutoff; dispersion and higher-order effects can arise	Compact, low cost and PCB-friendly; more radiation and less shielding than stripline/coax

2.2 Why the modes differ

TEM waves have both electric and magnetic fields entirely transverse to the direction of propagation. A two-conductor coaxial line supports a TEM field because a potential difference can be established between its inner and outer conductors. In a homogeneous dielectric, the TEM phase velocity is $c/\sqrt{\varepsilon r}$.

TE modes have no longitudinal electric-field component ($Ez = 0$), whereas TM modes have no longitudinal magnetic-field component ($Hz = 0$). A hollow single-conductor waveguide cannot support a TEM mode because a TEM solution requires two conductors with a non-zero voltage difference and a corresponding transverse electrostatic field.

Microstrip is quasi-TEM because its electric field occupies both the substrate and the air above it. The field therefore does not reside in one homogeneous dielectric, and the phase velocity is represented approximately using an effective relative permittivity εeff.

2.3 Weighted decision matrix

Scores use a 1–5 scale, where 5 is most suitable. The weights sum to 100%. The scores are engineering judgements for the stated low-power 2.4 GHz receiver-feed application and must be interpreted as a transparent decision aid rather than measured performance.

Criterion	Weight	Coaxial	Waveguide	Stripline	Microstrip
Operating-band suitability	20%	5	3	4	5
Loss performance	20%	4	5	4	3
Power handling	10%	4	5	3	2
Size and weight	10%	3	2	5	5
Cost	15%	4	2	4	5
Shielding / EMC	15%	5	5	4	2
Integration and connectors	10%	5	2	4	5
Weighted score (out of 5)	100%	4.35	3.55	4.00	3.85

Under these illustrative weights, coaxial line scores 4.35/5, waveguide 3.55/5, stripline 4.00/5 and microstrip 3.85/5. Coaxial line is selected because the application is a shielded feed between RF subsystems, where connector compatibility and EMI containment are important.

2.4 Sensitivity and alternative defence

If PCB integration and cost receive substantially greater weights, microstrip may rank first. If high-power handling and very low loss dominate at a much higher frequency, waveguide may become preferable. If shielding and connectorised interconnection remain dominant, coaxial line retains a strong ranking. The ranking therefore depends on application-specific weights and must not be treated as universal.

Design Decision 1: For the specified 2.4 GHz receiver feed, I select a coaxial line over microstrip because coaxial construction provides better shielding and convenient connectorised interconnection. The line supports TEM propagation without a TEM cutoff. The approximate TE11 limit calculated in Stage 3 is well above the selected band, although an exact modal limit would require a more detailed eigenmode analysis.

If frequency is doubled, conductor loss and connector/transition behaviour change and the risk of higher-order modes increases. If frequency is halved, the coax remains TEM, but wavelength increases and the electrical dimensions become smaller relative to wavelength. The alternative structure's ranking must be reassessed for the changed band.

3. STAGE 3 – MODE AND PROPAGATION MODEL

3.1 Given design parameters

Parameter	Symbol	Value
Centre frequency	f_0	2.400×10^9 Hz
Inner conductor radius	a	0.500 mm = 5.00×10^{-4} m
Inner radius of outer conductor	b	1.700 mm = 1.70×10^{-3} m
Relative permittivity	ε_r	2.25
Line length	L	0.50 m
Copper conductivity	σ	5.8×10^7 S/m
Dielectric loss tangent	$\tan\delta$	0.0005

3.2 Telegrapher's equations

For a uniform transmission line with per-unit-length resistance R, inductance L′, conductance G and capacitance C′, the time-domain telegrapher's equations are:

$$\partial V/\partial z = -R\, I - L'\, \partial I/\partial t$$

$$\partial I/\partial z = -G\, V - C'\, \partial V/\partial t$$

For sinusoidal steady state using the $e^{j\omega t}$ convention:

$$dV/dz = -(R + j\omega L')I; \quad dI/dz = -(G + j\omega C')V$$

$$\gamma = \alpha + j\beta = \sqrt{[(R + j\omega L')(G + j\omega C')]}$$

$$Z_0 = \sqrt{[(R + j\omega L')/(G + j\omega C')]}$$

For a low-loss line where $R \ll \omega L'$ and $G \ll \omega C'$, the approximate expressions are:

$$\alpha \approx R/(2Z_0) + GZ_0/2; \quad \beta \approx \omega\sqrt{(L'C')}$$

$$\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$$

3.3 Coaxial characteristic impedance – hand calculation

For a homogeneous dielectric-filled coaxial line, the lossless TEM characteristic impedance is:

$$Z_c = (60/\sqrt{\varepsilon_r})\ln(b/a)$$

Substitute a = 0.500 mm, b = 1.700 mm and $\varepsilon_r = 2.25$:

$$Z_c = (60/\sqrt{2.25})\ln(1.700/0.500)$$

$$Z_c = 40\ln(3.4) = 48.95\ \Omega \text{ (approximately)}$$

Result: $Z_c \approx 48.95\ \Omega$. This is close to the 50 Ω system reference but is not exactly 50 Ω. The ratio b/a or dielectric constant should be adjusted if an exact 50 Ω design is required.

3.4 Phase velocity, wavelength and phase constant

$$v_p = c/\sqrt{\varepsilon_r} = (2.99792458 \times 10^8)/1.5$$

$$v_p = 1.9986 \times 10^8 \text{ m/s}$$

$$\lambda = v_p/f_0 = (1.9986 \times 10^8)/(2.4 \times 10^9) = 0.08328 \text{ m}$$

$$\lambda \approx 83.28 \text{ mm}$$

$$\beta = 2\pi/\lambda \approx 75.45 \text{ rad/m}$$

For the ideal nondispersive TEM model, group velocity equals phase velocity: $v_g \approx v_p$. The group delay over 0.50 m is:

$$\tau_g = L/v_g = 0.50/(1.9986 \times 10^8) \approx 2.50 \text{ ns}$$

3.5 Approximate higher-order mode limit

The supplied MATLAB script uses the approximate estimate below for the first higher-order TE11 mode. It is used only as a preliminary screening calculation and is not an exact coaxial eigenvalue solution.

$$f_{TE11} \approx c/[\pi(a + b)\sqrt{\varepsilon_r}]$$

$$f_{TE11} \approx 2.9979 \times 10^8 / [\pi(0.0005 + 0.0017)(1.5)]$$

$$f_{TE11} \approx 28.9 \text{ GHz (approximate)}$$

The selected 2.350–2.450 GHz band is far below this estimate. A robust final design should still use manufacturer modal data or a validated eigenmode solver, and should consider the connector and feed transition.

3.6 TEM, TE and TM comparison

Property	TEM	TE	TM
Longitudinal fields	$E_z = 0$ and $H_z = 0$	$E_z = 0$; H_z may be non-zero	$H_z = 0$; E_z may be non-zero
Cutoff in hollow guide	Not supported in a hollow single-conductor guide; coax TEM has no cutoff	Cutoff generally exists	Cutoff generally exists
Dispersion	Ideal homogeneous TEM is nondispersive	Dispersive above cutoff	Dispersive above cutoff
Wave impedance	TEM impedance determined by line geometry/material	$Z_{TE} = \omega\mu/\beta$	$Z_{TM} = \beta/(\omega\varepsilon)$
Suitability here	Selected for the coaxial feed	Higher-order mode to avoid	Not the intended mode

3.7 Design decision 2

The intended dominant mode is TEM, which has no cutoff frequency. The approximate TE11 upper-mode estimate is 28.9 GHz; the selected 2.350–2.450 GHz band lies well below it. The dimensions are acceptable for a preliminary TEM model, subject to confirmation of exact characteristic impedance, connector transitions, and the validity of the approximate higher-mode limit.

4. STAGE 4 – LOSS EVALUATION AND LOSS-REDUCTION STRATEGY

4.1 Skin depth and surface resistance

$$\delta = \sqrt{[2/(\omega\mu\sigma)]} = \sqrt{[1/(\pi f\mu\sigma)]}$$

At f = 2.4 GHz, for copper with $\mu \approx \mu_0 = 4\pi \times 10^{-7}$ H/m and $\sigma = 5.8 \times 10^7$ S/m:

$$\delta \approx \sqrt{[1/(\pi \times 2.4 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7)]}$$

$$\delta \approx 1.35 \times 10^{-6} \text{ m} = 1.35 \ \mu\text{m}$$

$$R_s = \sqrt{(\pi f\mu/\sigma)} \approx 0.0128 \ \Omega \text{ per square}$$

4.2 Conductor attenuation – hand calculation

Using the approximate low-loss coaxial expression applied in the MATLAB model:

$$\alpha_c \approx R_s/(2Z_c) \times (1/a + 1/b) \quad \text{nepers/m}$$

Using $R_s \approx 0.01278 \ \Omega$, $Z_c \approx 48.95 \ \Omega$, a = 0.0005 m and b = 0.0017 m:

$$\alpha_c \approx [0.01278/(2 \times 48.95)] \times [1/0.0005 + 1/0.0017]$$

$$\alpha_c \approx 0.338 \text{ Np/m (approximately)}$$

$$A_c = 8.686 \ \alpha_c \approx 2.93 \text{ dB/m}$$

Over L = 0.50 m, the approximate conductor loss is $A_cL \approx 1.47$ dB.

4.3 Dielectric attenuation – hand calculation

$$\alpha_d \approx (\beta \tan\delta)/2$$

Using $\beta \approx 75.45$ rad/m and $\tan\delta = 0.0005$:

$$\alpha_d \approx (75.45 \times 0.0005)/2 = 0.01886 \text{ Np/m}$$

$$A_d = 8.686 \ \alpha_d \approx 0.164 \text{ dB/m}$$

Over 0.50 m, the approximate dielectric loss is $A_dL \approx 0.082$ dB.

4.4 Total distributed loss and contribution

$$A_{total} \approx A_c + A_d \approx 2.93 + 0.164 = 3.10 \text{ dB/m}$$

$$\text{Loss over } 0.50 \text{ m} \approx 3.10 \times 0.50 = 1.55 \text{ dB}$$

Using the rounded values, conductor loss contributes about 94.7% of the distributed loss and dielectric loss about 5.3%. The exact MATLAB summary may differ slightly because it uses the unrounded values.

Loss component	Initial model (dB/m)	Improved design (dB/m)	Share of initial distributed loss	Remarks
Conductor loss	Insert MATLAB value	To be determined	Approximately 95% in preliminary model	Dominant mechanism; depends on conductivity, roughness and geometry
Dielectric loss	Insert MATLAB value	To be determined	Approximately 5% in preliminary model	Depends on loss tangent and dielectric properties
Radiation loss	Not separately modelled	Requires layout/connector estimate or measurement	Not quantified by this script	Coax shielding normally reduces radiation if properly terminated

Connector/transition loss	Not separately modelled	Requires datasheet or measurement	Not quantified by this script	Must be added for a complete practical link budget
Mismatch loss	Calculate separately from Γ	Calculate after matching	Load-dependent	Do not combine with distributed loss without stating the convention
Total	Distributed model only	Complete after redesign	—	Current script does not quantify every requested loss mechanism

4.5 Parametric trends

- Frequency: conductor surface resistance generally increases approximately with √f; dielectric attenuation in the low-loss approximation increases with frequency.
- Conductor material: higher conductivity reduces surface resistance and conductor loss, although cost, corrosion resistance and mechanical properties must also be considered.
- Surface roughness: rough conductor surfaces can increase RF loss when roughness is significant relative to skin depth.
- Dielectric loss tangent: reducing tanδ reduces dielectric attenuation approximately in direct proportion in the low-loss model.

4.6 Proposed loss-reduction strategy

A defensible first improvement is to use a lower-loss coaxial cable with a verified manufacturer attenuation specification at 2.4 GHz, low-loss dielectric, suitable conductor/plating and correctly installed connectors. The improvement must be quantified using datasheet attenuation or measured insertion loss. The supplied script does not include a complete material/roughness model, so it cannot honestly establish a precise improved-design dB value on its own.

Design Decision 3: The preliminary model indicates conductor loss is dominant. The recommended improvement is a lower-attenuation coaxial cable and careful connector installation. The final before/after attenuation values must be entered from a verified datasheet or independent measurement; they must not be invented.

5. STAGE 5 – NETWORK MODEL, MATCHING AND S-PARAMETERS

5.1 Equivalent voltage and current

For TEM transmission lines, voltage and current can be defined using transverse field integrals and conductor current. For non-TEM waveguides, a unique voltage and current are not naturally defined in the same way because the field structure is not a uniform transverse TEM field. Equivalent voltage/current quantities may be defined for a selected mode and normalization, but the convention must be stated.

5.2 Normalized wave variables

$$a = (V + Z_0 I)/(2\sqrt{\mathrm{Re}\{Z_0\}})$$

$$b = (V - Z_0^* I)/(2\sqrt{\mathrm{Re}\{Z_0\}})$$

For a real reference impedance Z_0, these power-wave definitions reduce to the familiar incident and reflected voltage-wave relationships. For a two-port network, [b] = [S][a], where S11 and S22 are input/output reflection coefficients and S21 and S12 are forward/reverse transmission coefficients.

5.3 ABCD matrix of a uniform line

$$[A\ \ B;\ C\ \ D] = [\cosh(\gamma L),\ Z_c \sinh(\gamma L);\ \sinh(\gamma L)/Z_c,\ \cosh(\gamma L)]$$

The MATLAB script uses the lossy propagation constant $\gamma = \alpha + j\beta$, with α approximated by the sum of conductor and dielectric attenuation in nepers per metre. For equal real reference impedances Zref at both ports, the ABCD-to-S conversion is:

$$\Delta = A + B/Z_{ref} + C\,Z_{ref} + D$$

$$S11 = (A + B/Z_{ref} - C\,Z_{ref} - D)/\Delta$$

$$S21 = 2/\Delta$$

$$S12 = 2(AD - BC)/\Delta$$

$$S22 = (-A + B/Z_{ref} - C\,Z_{ref} + D)/\Delta$$

5.4 Reflection coefficient, VSWR and return loss – hand calculation

For the preliminary line impedance Zc = 48.95 Ω and a 75 Ω resistive load:

$$\Gamma L = (Z_L - Z_c)/(Z_L + Z_c) = (75 - 48.95)/(75 + 48.95)$$

$$|\Gamma L| \approx 0.210$$

$$VSWR = (1 + |\Gamma L|)/(1 - |\Gamma L|) \approx 1.53$$

$$RL = -20 \log_{10}|\Gamma L| \approx 13.55\ \mathrm{dB}$$

These values describe the load mismatch referenced to the calculated line impedance. The uniform-line S-parameters in the MATLAB script use equal 50 Ω reference impedances and represent a separate two-port line model.

5.5 Quarter-wave transformer matching

For a real load connected to a line, a quarter-wave transformer with characteristic impedance $Z_t = \sqrt{(Z_c Z_L)}$ matches the load to the line at the design frequency, provided the transformer is lossless and exactly one quarter wavelength long.

$$Z_t = \sqrt{(Z_c Z_L)} = \sqrt{(48.95 \times 75)} \approx 60.6\ \Omega$$

$$Lt = \lambda/4 = 83.28/4 \approx 20.82 \text{ mm}$$

At the centre frequency, the ideal quarter-wave section transforms the 75 Ω load to approximately Zc. The match degrades away from the centre frequency, and actual conductor/dielectric losses and discontinuities affect the result. The MATLAB script computes the transformer input impedance and reflection across the selected band.

5.6 S-parameter interpretation

Property	Expected interpretation	How to verify
Reciprocity	For a passive reciprocal line, S21 ≈ S12	Compare the complex S21 and S12 values
Symmetry	A uniform line with identical port reference planes has S11 ≈ S22	Compare S11 and S22 across the band
Losslessness	For an ideal lossless network, S†S = I	The current lossy line should not be described as lossless
Matching	Good matching means small |S11| and large return loss	Inspect S11 magnitude and calculate −20 log10|S11|
Insertion loss	Insertion loss is −20 log10|S21| for matched equal-reference ports	Use S21 magnitude and state the reference conditions

5.7 Network summary table

Parameter	Before matching	After matching	Target	Interpretation
Characteristic impedance	Calculated Zc ≈ 48.95 Ω	Line remains the same unless redesigned	Nominal 50 Ω	Close to, but not exactly, 50 Ω
Reflection coefficient	|Γ| ≈ 0.210 at 75 Ω load	Ideally near zero at centre frequency	Small |Γ|	Verify from the MATLAB sweep
VSWR	≈ 1.53	Ideally 1.00 at centre frequency	≤ 1.92	10 dB return-loss target equivalent
Return loss	≈ 13.55 dB	High at centre for ideal match	≥ 10 dB	Verify over entire band
Insertion loss	From S21 output	Matched system needs separate cascade model	Project-specific limit	Do not confuse with load mismatch loss
Reciprocity	Check S21 versus S12	Expected for reciprocal passive line	Difference near numerical precision	Report numerical error

5.8 Output spaces

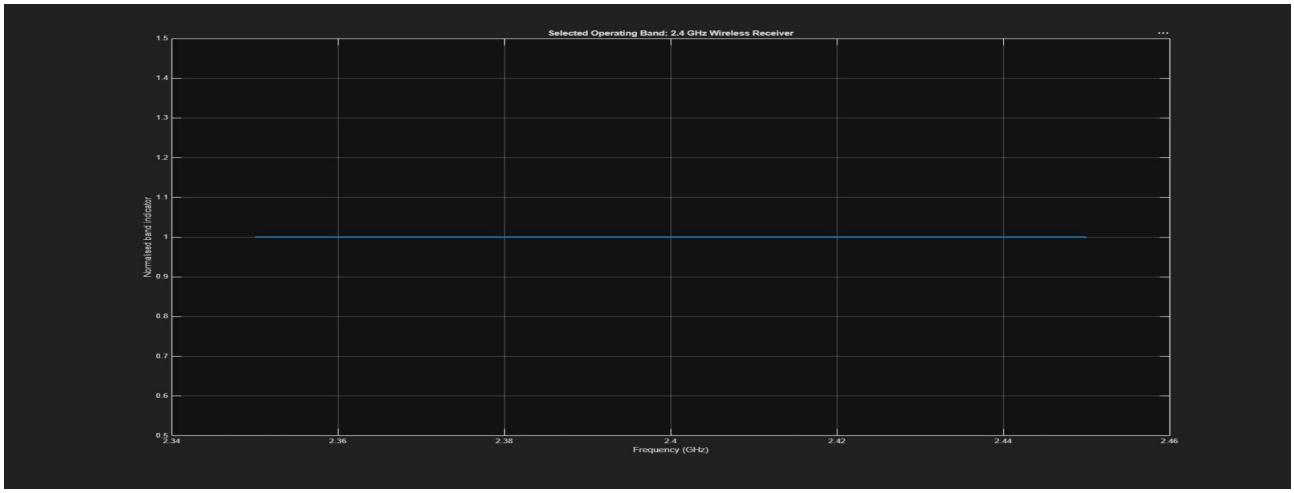


Figure 1. Selected Operating Band

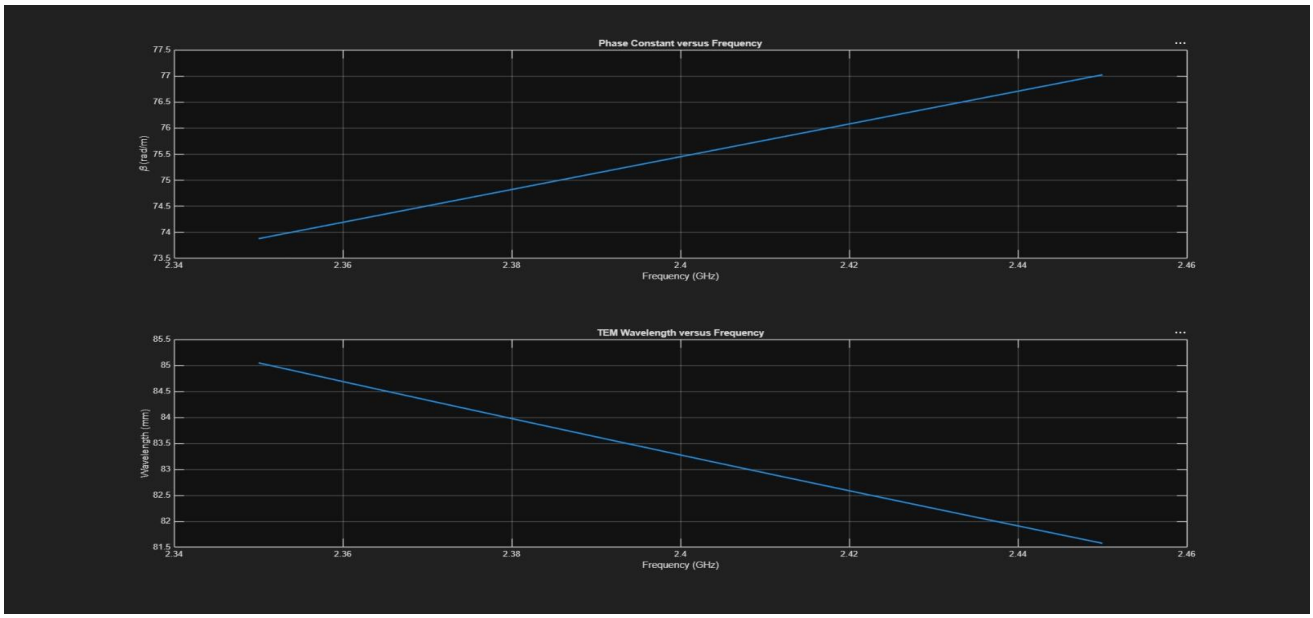


Figure 2. Phase Constant and TEM Wavelength versus Frequency

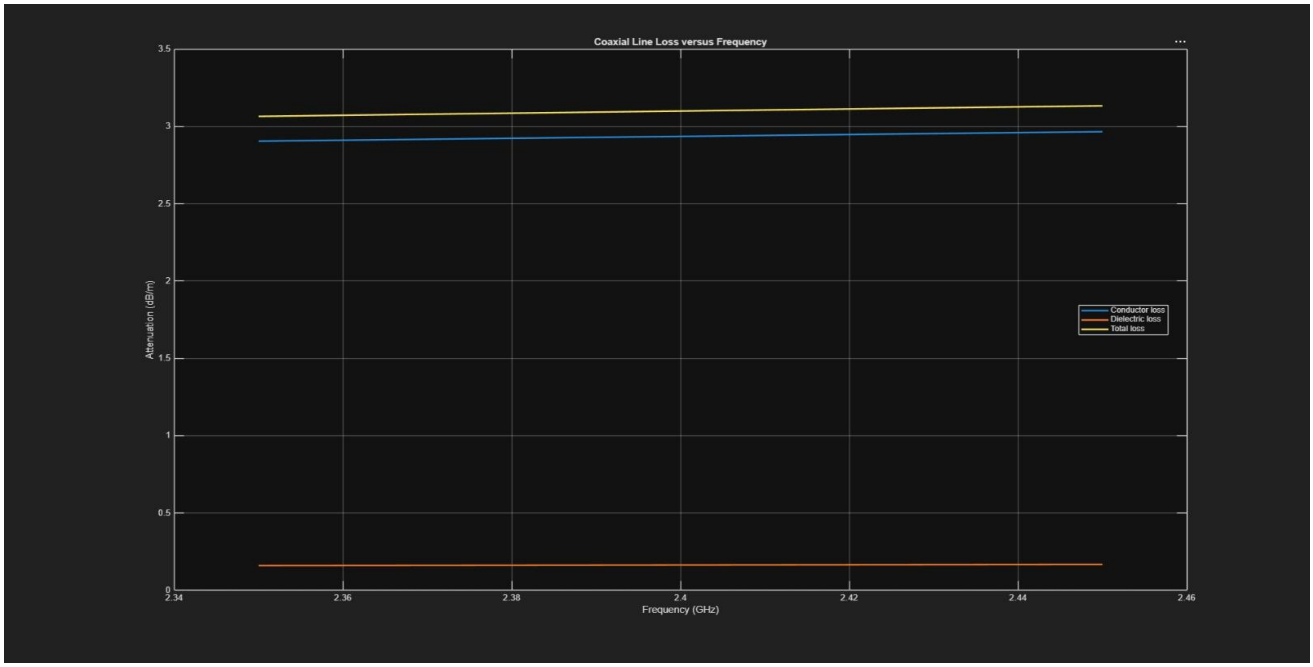


Figure 3. Conductor, Dielectric and Total Attenuation versus Frequency

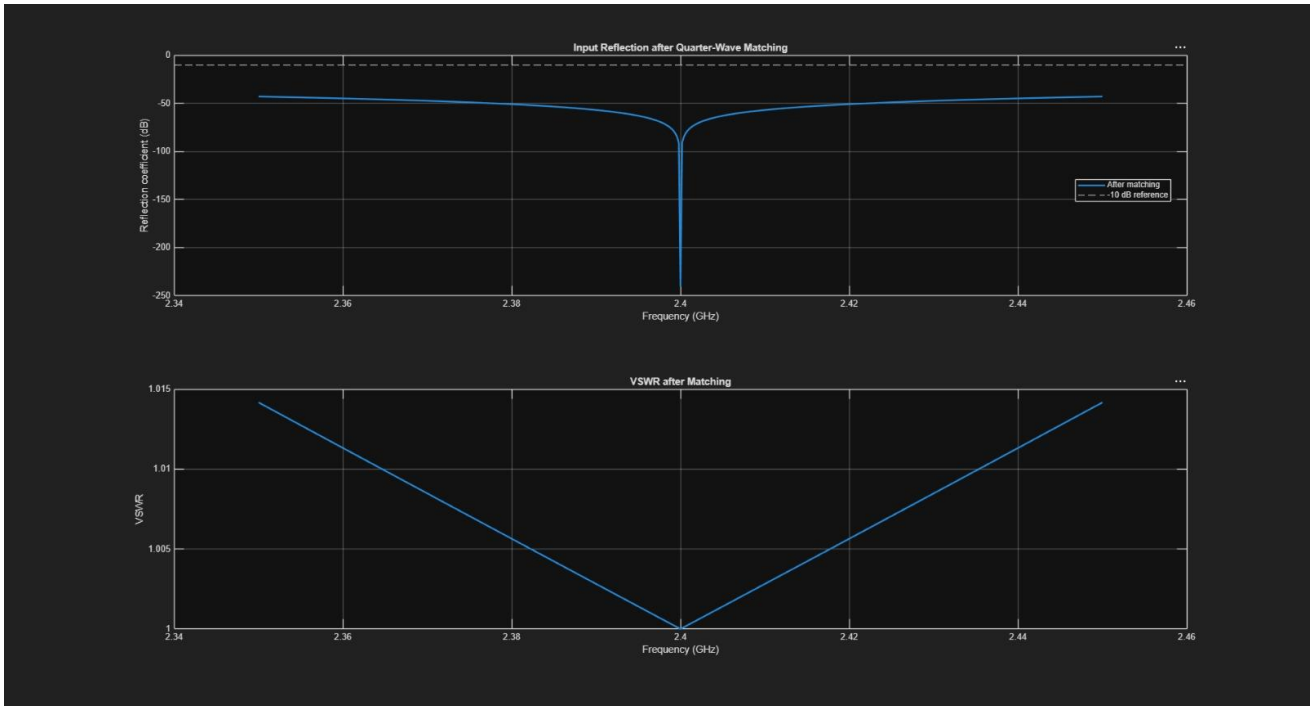


Figure 4. Reflection and VSWR after Quarter-Wave Matching

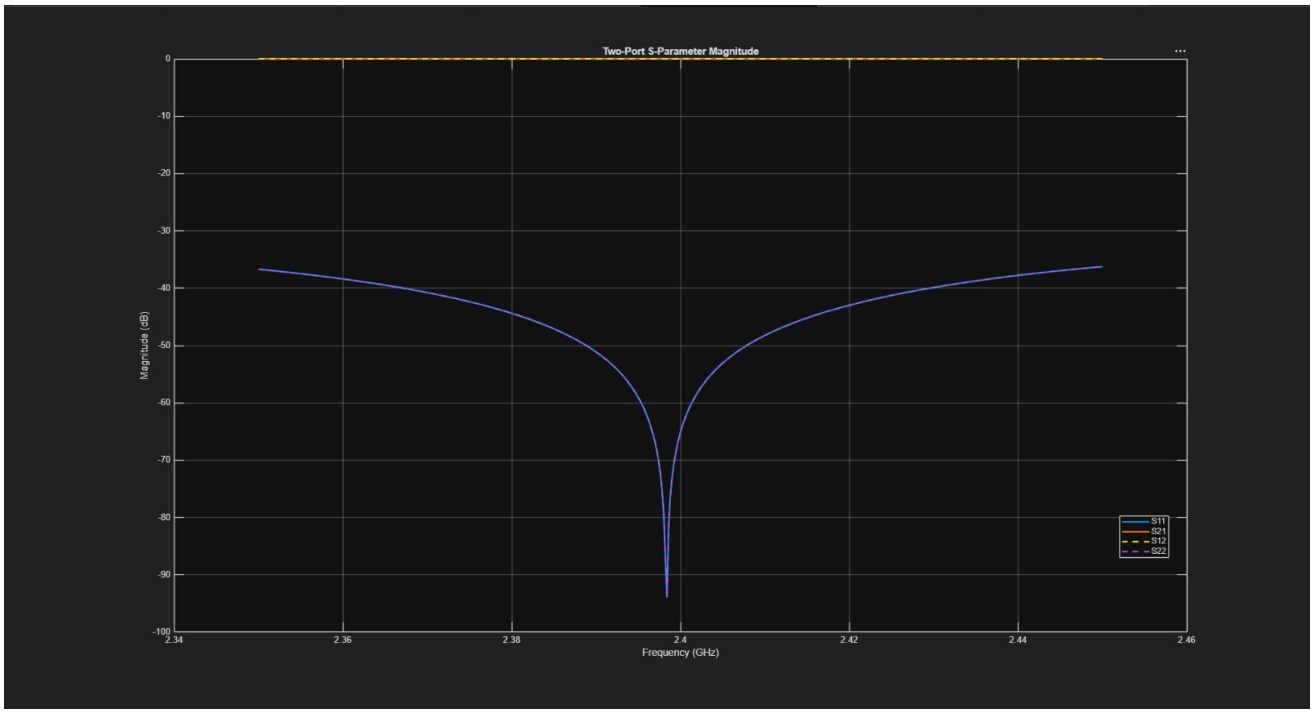


Figure 5. Two-Port S-Parameter Magnitude

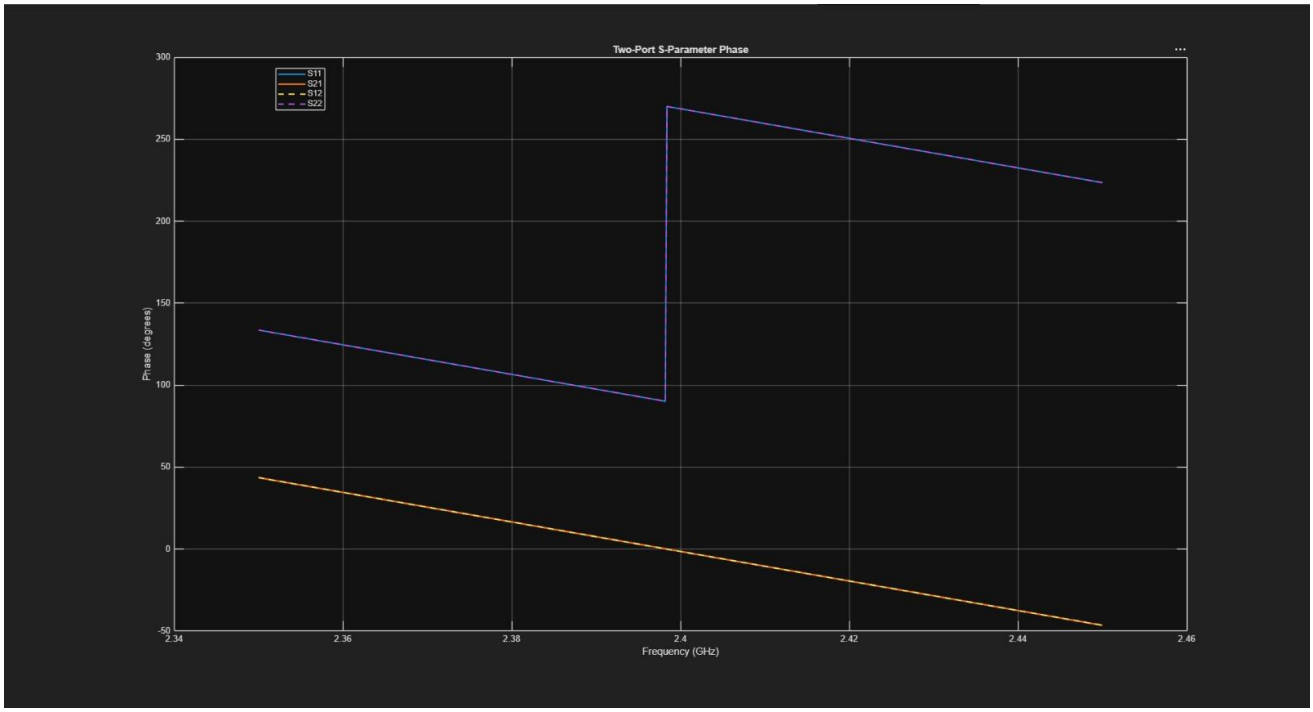


Figure 6. Two-Port S-Parameter Phase

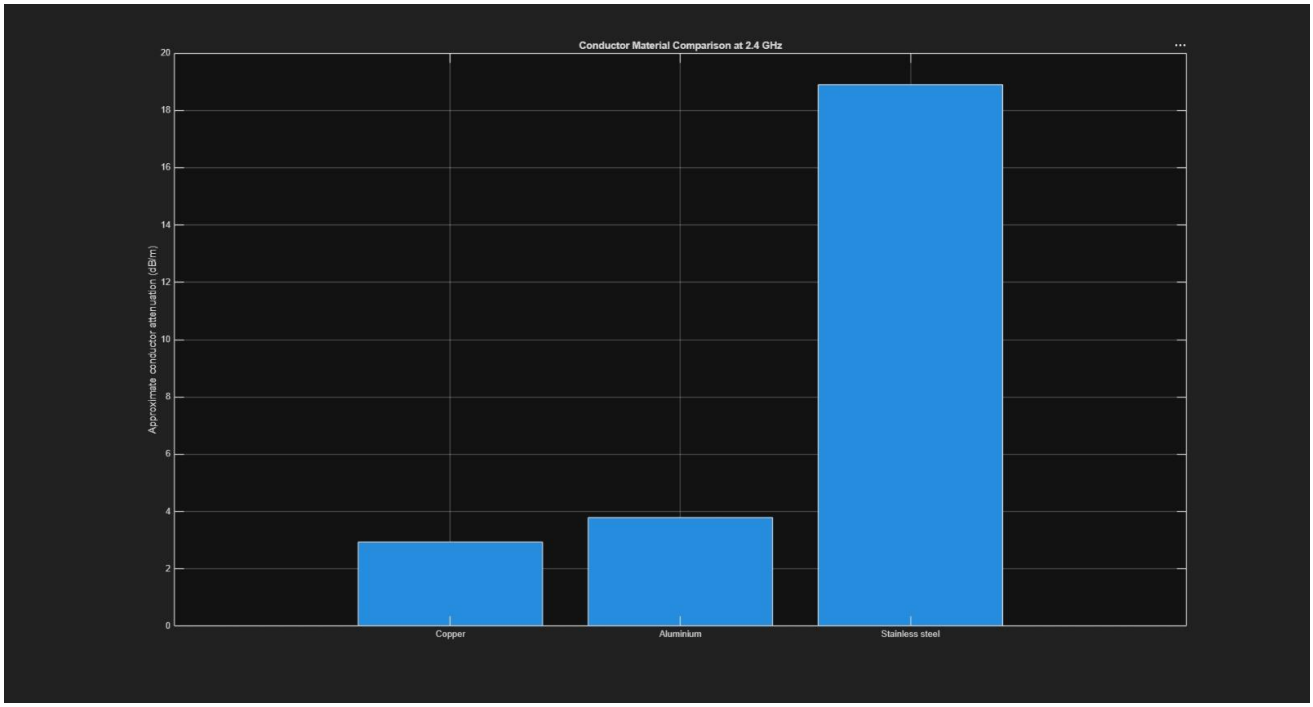


Figure 7. Conductor Material Comparison

6. STAGE 6 – VALIDATION, CRITIQUE AND TOLERANCE ANALYSIS

6.1 Validation approach

The initial analytical checks are the coaxial impedance formula, TEM velocity and wavelength, approximate skin depth and surface resistance, low-loss attenuation equations, and the reflection/VSWR relations. MATLAB is used to evaluate the frequency sweep and to cross-check the arithmetic. This is analytical/software verification; it is not an experimental measurement.

6.2 Tolerance estimate for dimensions

The characteristic impedance depends on the ratio b/a. For small independent dimension changes, a first-order differential estimate is:

$$dZc \approx (60/\sqrt{\varepsilon r})\,(db/b - da/a)$$

For an illustrative ±5% tolerance in a and b, the worst first-order ratio change occurs when one dimension increases by 5% while the other decreases by 5%. The logarithmic ratio change is approximately $\ln(1.05/0.95) = 0.1001$.

$$|\Delta Zc| \approx (60/\sqrt{2.25}) \times 0.1001 \approx 4.00\ \Omega$$

This is a conservative first-order estimate of approximately ±4 Ω around the nominal calculated impedance, depending on the direction of the dimensional errors. Exact worst-case values should be calculated by evaluating the impedance formula at the tolerance extremes.

6.3 Assumptions and limitations

- Idealized homogeneous dielectric: real cables can have non-uniform materials and frequency-dependent permittivity.
- Smooth conductor: the basic model does not explicitly model surface roughness or plating thickness.
- Approximate TE11 cutoff: the expression in the code is a screening estimate, not a full eigenmode solution.
- Uniform line: connector, bend, launch and transition discontinuities are not included in the line's distributed-loss calculation.
- No hardware measurement: plotted results are calculated outputs, not VNA measurements.
- Matching and S-parameter models are separate: the plotted S-parameters describe the uniform line, while the quarter-wave transformer calculation is analysed separately.

6.4 EMC and safety test plan

1. Inspect the coaxial cable, connector condition, grounding and outer-conductor continuity.
2. Use a calibrated vector network analyser (VNA), where available, to measure S11 and S21 across 2.350–2.450 GHz.
3. Perform a return-loss/VSWR check using a suitable 50 Ω termination and compare the measured results with the calculated model.
4. Check for unwanted radiation or coupling around connectors and transitions using an appropriate EMC test arrangement.
5. Operate only within the rated power of the cable, connectors, instruments and receiver; follow laboratory RF exposure and electrical safety procedures.
6. Record calibration method, cable length, connector type, measurement uncertainty and environmental conditions.

6.5 Specification verification table

Requirement	Specified	Initial design	Final design (worst case)	Satisfied?
Operating band	2.350–2.450 GHz	Within analysis sweep	To be verified after tolerances	Pending
Dominant mode	TEM	TEM intended; TE11 estimate ≈ 28.9 GHz	Confirm with detailed modal data	Preliminary
Characteristic impedance	50 Ω nominal	48.95 Ω calculated	Adjust dimensions/material if exact 50 Ω required	Not exact
Distributed loss	≤ 1.6 dB / 0.5 m provisional	Approx. 1.55 dB from rounded analytical values; use MATLAB result	Use verified cable data/measurement	Provisional
Return loss	≥ 10 dB target	≈ 13.55 dB for 75 Ω load relative to line Zc	Verify matched system over band	Initial load meets target
VSWR	≤ 1.92 target	≈ 1.53 for 75 Ω load	Verify matched system over band	Initial load meets target
EMI shielding/isolation	Set from system requirement	Not quantified by script	EMC measurement required	Pending
Power/size	Set from selected hardware	L = 0.50 m model; power not rated	Confirm datasheets	Pending

7. FINAL ENGINEERING DECISION

7.1 Initial design decision

Decision: NO for unconditional adoption as a fully verified engineering design. The preliminary coaxial geometry supports the intended TEM operating principle and its calculated impedance is close to 50 Ω. However, the impedance is not exactly 50 Ω, the higher-order-mode limit is approximate, and the complete practical loss budget, shielding performance, connector effects and tolerance verification have not been established by the MATLAB script alone.

7.2 Improved design decision

Decision: CONDITIONAL YES after the following checks are completed: adjust the coaxial geometry or select a nominal 50 Ω commercial cable; use manufacturer attenuation data or measured S21 to establish the true loss budget; validate S11/VSWR over the full band; and verify connector, shielding, power and tolerance requirements. If these checks do not meet the specified targets, the final decision must remain NO until the remaining non-compliance is corrected.

7.3 Further improvement

A commercially specified 50 Ω coaxial cable with documented attenuation at 2.4 GHz is the most direct practical alternative to the illustrative custom geometry. It improves reproducibility and provides traceable performance data. The resulting loss and return-loss values must be entered from a verified datasheet or VNA measurement; they are not assumed here.

Conclusion

This report develops a preliminary coaxial TEM transmission-line model for a 2.4 GHz wireless receiver feed. The analytical results establish the impedance, propagation velocity, wavelength, approximate mode limit, conductor/dielectric loss and basic load-matching behaviour. MATLAB provides a frequency-domain calculation and plotting framework. Final adoption requires completion of the practical loss budget, verification of a nominal 50 Ω implementation, modal confirmation, and measurement-based validation of the connectors and assembled feed.

APPENDIX – MATLAB CODE

The code below is the fixed version supplied for the analysis. It prints scalar centre-frequency loss results only once and generates seven figures. Paste any updated final code here if you change the model before submission.

```
%% ECA13 - MICROWAVE ENGINEERING
% CO1 Application-Based Assessment
% Application: 2.4 GHz Wireless Receiver Feed
% Structure: Coaxial Transmission Line
% Standard MATLAB functions only

clear;
clc;
close all;
format short g;

%% 1. DESIGN SPECIFICATIONS
c = 299792458;
mu0 = 4*pi*1e-7;
eps0 = 1/(mu0*c^2);

f0 = 2.4e9;
BW = 100e6;
f1 = f0 - BW/2;
f2 = f0 + BW/2;
f = linspace(f1,f2,501);
w = 2*pi*f;

Zref = 50;
er = 2.25;
tanD = 0.0005;
sigma = 5.8e7;

a = 0.5e-3;
b = 1.7e-3;
L = 0.5;
ZL = 75;

idx0 = ceil(numel(f)/2);

%% 2. CHARACTERISTIC IMPEDANCE
Zc = 60/sqrt(er)*log(b/a);

%% 3. TEM PROPAGATION
vp = c/sqrt(er);
beta = 2*pi*f/vp;
lambda = vp./f;
vg = vp;
tau = L/vg;

%% 4. APPROXIMATE HIGHER-ORDER MODE LIMIT
% Approximate TE11 estimate; not an exact eigenmode solution.
fTE11 = c/(pi*(a+b)*sqrt(er));

%% 5. CONDUCTOR AND DIELECTRIC LOSS
delta = sqrt(2./(w*mu0*sigma));
Rs = sqrt(w*mu0/(2*sigma));

alphaC = Rs./(2*Zc)*(1/a + 1/b);
alphaD = beta*tanD/2;

conv = 20/log(10);
Ac = conv*alphaC;
Ad = conv*alphaD;
At = Ac + Ad;

Ac0 = Ac(idx0);
Ad0 = Ad(idx0);
At0 = At(idx0);

lossC = Ac0*L;
```

```
lossD = Ad0*L;
lossTotal = At0*L;

%% 6. LOAD MATCHING
GammaL = (ZL-Zc)/(ZL+Zc);
VSWR_before = (1+abs(GammaL))/(1-abs(GammaL));
RL_before = -20*log10(max(abs(GammaL),1e-12));

Zt = sqrt(Zc*ZL);
lambda0 = vp/f0;
Lt = lambda0/4;

betaT = 2*pi*f/vp;
Zin_after = Zt.*(ZL + 1j*Zt.*tan(betaT*Lt)) ./ ...
   (Zt + 1j*ZL.*tan(betaT*Lt));

Gamma_after = (Zin_after-Zc)./(Zin_after+Zc);
VSWR_after = (1+abs(Gamma_after))./ ...
   max(1-abs(Gamma_after),1e-12);
RL_after = -20*log10(max(abs(Gamma_after),1e-12));

%% 7. ABCD MATRIX AND S-PARAMETERS
% Uniform line; equal reference impedance at both ports.
gammaLine = alphaC + alphaD + 1j*beta;

A = cosh(gammaLine*L);
B = Zc.*sinh(gammaLine*L);
C = sinh(gammaLine*L)./Zc;
D = A;

den = A + B/Zref + C*Zref + D;

S11 = (A + B/Zref - C*Zref - D)./den;
S21 = 2./den;
S12 = 2*(A.*D-B.*C)./den;
S22 = (-A + B/Zref - C*Zref + D)./den;

%% 8. PRINT RESULTS ONCE ONLY
fprintf('\n============================================\n');
fprintf('MICROWAVE ENGINEERING - DESIGN SUMMARY\n');
fprintf('============================================\n');

fprintf('\nDESIGN SPECIFICATIONS\n');
fprintf('Operating frequency       : %.3f GHz\n',f0/1e9);
fprintf('Bandwidth                 : %.1f MHz\n',BW/1e6);
fprintf('Line length               : %.3f m\n',L);
fprintf('Inner conductor radius    : %.3f mm\n',a*1e3);
fprintf('Outer conductor radius    : %.3f mm\n',b*1e3);
fprintf('Relative permittivity     : %.3f\n',er);
fprintf('Characteristic impedance  : %.3f ohms\n',Zc);

fprintf('\nTEM PROPAGATION\n');
fprintf('Phase velocity            : %.4e m/s\n',vp);
fprintf('Wavelength                : %.3f mm\n',lambda(idx0)*1e3);
fprintf('Phase constant            : %.3f rad/m\n',beta(idx0));
fprintf('Group velocity            : %.4e m/s\n',vg);
fprintf('Group delay               : %.3f ns\n',tau*1e9);

fprintf('\nMODE LIMIT\n');
fprintf('TEM cutoff                : None\n');
fprintf('Approximate TE11 estimate : %.3f GHz\n',fTE11/1e9);

fprintf('\nLOSS ANALYSIS AT CENTRE FREQUENCY\n');
fprintf('Skin depth                : %.3f micrometres\n',delta(idx0)*1e6);
fprintf('Surface resistance        : %.6f ohms\n',Rs(idx0));
fprintf('Conductor attenuation     : %.5f dB/m\n',Ac0);
fprintf('Dielectric attenuation    : %.5f dB/m\n',Ad0);
fprintf('Total attenuation         : %.5f dB/m\n',At0);
fprintf('Conductor loss over line  : %.5f dB\n',lossC);
fprintf('Dielectric loss over line : %.5f dB\n',lossD);
fprintf('Total distributed loss    : %.5f dB\n',lossTotal);
fprintf('Conductor loss share      : %.2f %%\n',100*Ac0/At0);
fprintf('Dielectric loss share     : %.2f %%\n',100*Ad0/At0);
```

```
fprintf('\nMATCHING ANALYSIS\n');
fprintf('Load impedance            : %.2f ohms\n',ZL);
fprintf('Transformer impedance     : %.3f ohms\n',Zt);
fprintf('Quarter-wave length       : %.3f mm\n',Lt*1e3);
fprintf('Before matching |Gamma|   : %.5f\n',abs(GammaL));
fprintf('Before matching VSWR      : %.4f\n',VSWR_before);
fprintf('Before matching return loss: %.3f dB\n',RL_before);
fprintf('After matching |Gamma|    : %.6f\n',abs(Gamma_after(idx0)));
fprintf('After matching VSWR       : %.4f\n',VSWR_after(idx0));
fprintf('After matching return loss: %.3f dB\n',RL_after(idx0));

fprintf('\nTWO-PORT NETWORK AT CENTRE FREQUENCY\n');
fprintf('|S11|                    : %.6f\n',abs(S11(idx0)));
fprintf('|S21|                    : %.6f\n',abs(S21(idx0)));
fprintf('|S12|                    : %.6f\n',abs(S12(idx0)));
fprintf('|S22|                    : %.6f\n',abs(S22(idx0)));
fprintf('Insertion loss from S21   : %.5f dB\n', ...
   -20*log10(max(abs(S21(idx0)),1e-12)));
fprintf('Reciprocity error         : %.3e\n', ...
   abs(S21(idx0)-S12(idx0)));

%% 9. FIGURE 1 - OPERATING BAND
figure('Name','Figure 1 - Operating Band');
plot(f/1e9,ones(size(f)),'LineWidth',1.8);
grid on;
xlabel('Frequency (GHz)');
ylabel('Band indicator');
title('Selected Operating Band');
ylim([0.5 1.5]);

%% 10. FIGURE 2 - PROPAGATION PARAMETERS
figure('Name','Figure 2 - Propagation');
subplot(2,1,1);
plot(f/1e9,beta,'LineWidth',1.8);
grid on;
xlabel('Frequency (GHz)');
ylabel('Phase constant (rad/m)');
title('Phase Constant versus Frequency');

subplot(2,1,2);
plot(f/1e9,lambda*1e3,'LineWidth',1.8);
grid on;
xlabel('Frequency (GHz)');
ylabel('Wavelength (mm)');
title('TEM Wavelength versus Frequency');

%% 11. FIGURE 3 - LOSS VERSUS FREQUENCY
figure('Name','Figure 3 - Loss');
plot(f/1e9,Ac,'LineWidth',1.8);
hold on;
plot(f/1e9,Ad,'LineWidth',1.8);
plot(f/1e9,At,'LineWidth',1.8);
grid on;
xlabel('Frequency (GHz)');
ylabel('Attenuation (dB/m)');
title('Coaxial Transmission Line Loss');
legend('Conductor','Dielectric','Total','Location','best');

%% 12. FIGURE 4 - MATCHING
figure('Name','Figure 4 - Matching');
subplot(2,1,1);
plot(f/1e9,20*log10(max(abs(Gamma_after),1e-12)), ...
   'LineWidth',1.8);
hold on;
yline(-10,'--');
grid on;
xlabel('Frequency (GHz)');
ylabel('Reflection (dB)');
title('Reflection after Quarter-Wave Matching');
legend('Matched load','-10 dB reference','Location','best');

subplot(2,1,2);
plot(f/1e9,VSWR_after,'LineWidth',1.8);
grid on;
```

```
xlabel('Frequency (GHz)');
ylabel('VSWR');
title('VSWR after Matching');

%% 13. FIGURE 5 - S-PARAMETER MAGNITUDE
figure('Name','Figure 5 - S Parameter Magnitude');
plot(f/1e9,20*log10(max(abs(S11),1e-12)),'LineWidth',1.5);
hold on;
plot(f/1e9,20*log10(max(abs(S21),1e-12)),'LineWidth',1.5);
plot(f/1e9,20*log10(max(abs(S12),1e-12)),'--','LineWidth',1.5);
plot(f/1e9,20*log10(max(abs(S22),1e-12)),'--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Two-Port S-Parameter Magnitude');
legend('S11','S21','S12','S22','Location','best');

%% 14. FIGURE 6 - S-PARAMETER PHASE
figure('Name','Figure 6 - S Parameter Phase');
plot(f/1e9,unwrap(angle(S11))*180/pi,'LineWidth',1.5);
hold on;
plot(f/1e9,unwrap(angle(S21))*180/pi,'LineWidth',1.5);
plot(f/1e9,unwrap(angle(S12))*180/pi,'--','LineWidth',1.5);
plot(f/1e9,unwrap(angle(S22))*180/pi,'--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Phase (degrees)');
title('Two-Port S-Parameter Phase');
legend('S11','S21','S12','S22','Location','best');

%% 15. FIGURE 7 - CONDUCTOR MATERIAL COMPARISON
materials = {'Copper','Aluminium','Stainless steel'};
conductivity = [5.8e7,3.5e7,1.4e6];

RsMat = sqrt(pi*f0*mu0./conductivity);
AcMat = RsMat/(2*Zc)*(1/a+1/b)*conv;

figure('Name','Figure 7 - Material Comparison');
bar(AcMat);
grid on;
set(gca,'XTickLabel',materials);
ylabel('Approximate conductor attenuation (dB/m)');
title('Conductor Material Comparison at 2.4 GHz');

%% 16. SUMMARY TABLE
Parameter = ["Frequency";"Characteristic impedance"; ...
   "Phase velocity";"Wavelength";"Conductor attenuation"; ...
   "Dielectric attenuation";"Total attenuation"; ...
   "Return loss before matching";"Return loss after matching"];

Value = [f0/1e9;Zc;vp;lambda(idx0)*1e3;Ac0;Ad0;At0; ...
   RL_before;RL_after(idx0)];

Unit = ["GHz";"ohms";"m/s";"mm";"dB/m";"dB/m";"dB/m";"dB";"dB"];

Summary = table(Parameter,Value,Unit);

fprintf('\nSUMMARY TABLE\n');
disp(Summary);

fprintf('\nAnalysis completed successfully.\n');
fprintf('Seven figures generated.\n');
```

REFERENCES

1. D. M. Pozar, Microwave Engineering, 4th ed., John Wiley & Sons, 2012.
2. C. A. Balanis, Advanced Engineering Electromagnetics, 2nd ed., John Wiley & Sons, 2012.
3. D. K. Cheng, Field and Wave Electromagnetics, 2nd ed., Addison-Wesley, 1989.
4. MATLAB documentation, MathWorks. Standard functions used include plot, subplot, table, cosh, sinh, angle and unwrap. Consult the installed MATLAB documentation for version-specific details.
5. For a final practical design, cite the datasheet of the exact 50 Ω coaxial cable and connectors used, including attenuation and frequency ratings.